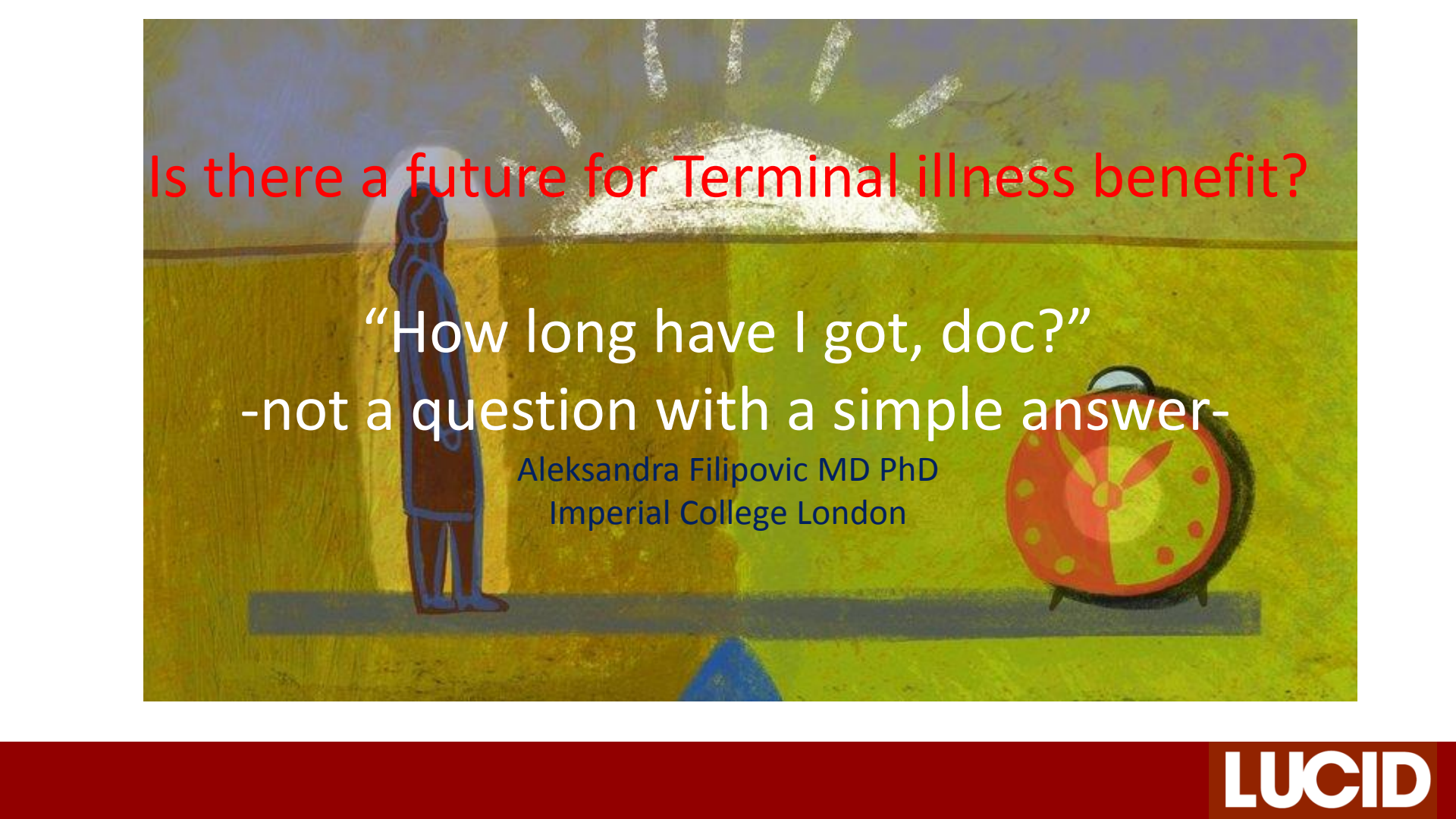




Is there a future for Terminal illness benefit?

“How long have I got, doc?”
-not a question with a simple answer-

Aleksandra Filipovic MD PhD
Imperial College London

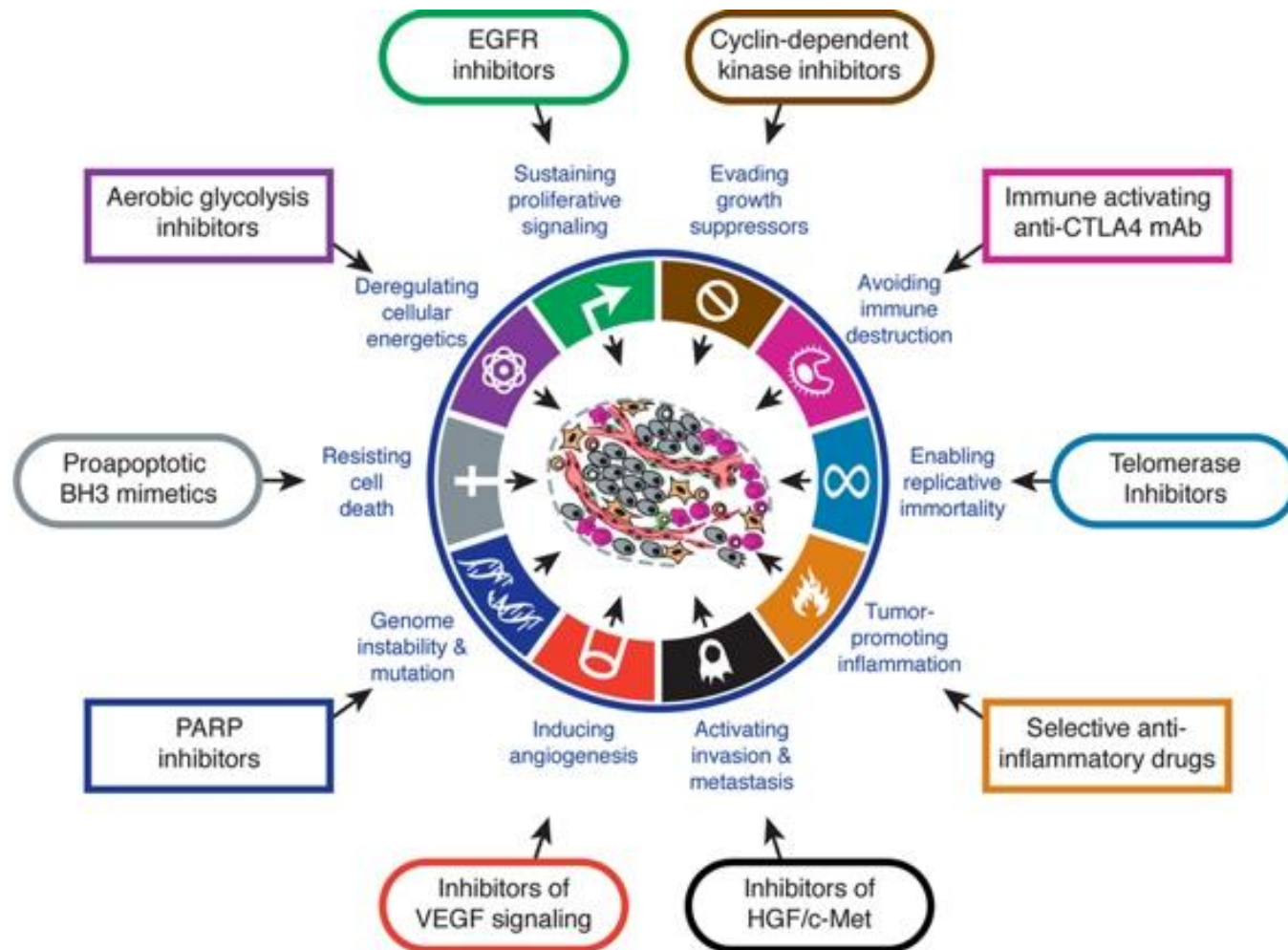
Agenda

- **Challenges Oncologists face in accurately predicting survival**
- New Therapies
 - Approved / non-approved
 - Personalised / targeted & Immunotherapy
- Cancers with greatest prognostic improvements
- Costs of clinical trials and gaining approval
 - Self funded
 - Postcode prescribing – is it a post code lottery?



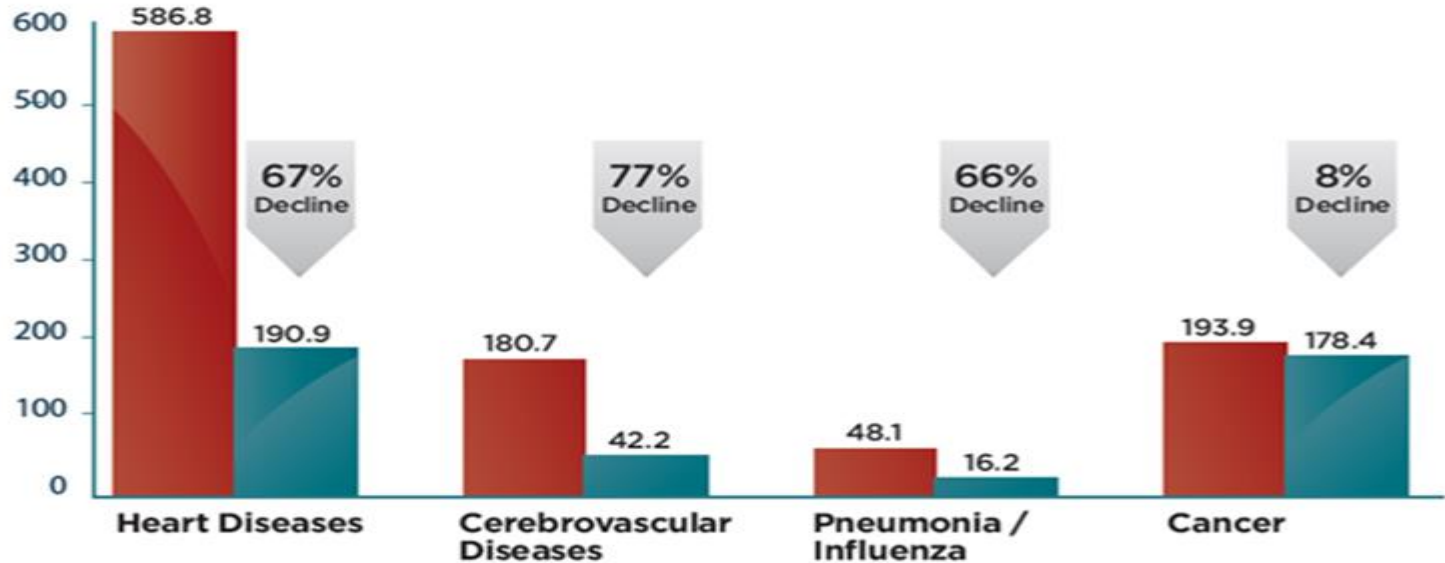
- Cancer is not one disease but refers to a large number of diseases.
- One defining feature of cancer is the rapid creation of abnormal cells that grow beyond their usual boundaries, and which can then spread to other organs.
- New cases of cancer are estimated to jump from 11.3 million in 2007 to 22 million in 2035. We now have better screening and diagnostic methods as well as a more educated public.
- Of all cancers, 30–40 % are preventable.

Cancer is a heterogeneous disease with complex molecular characteristics



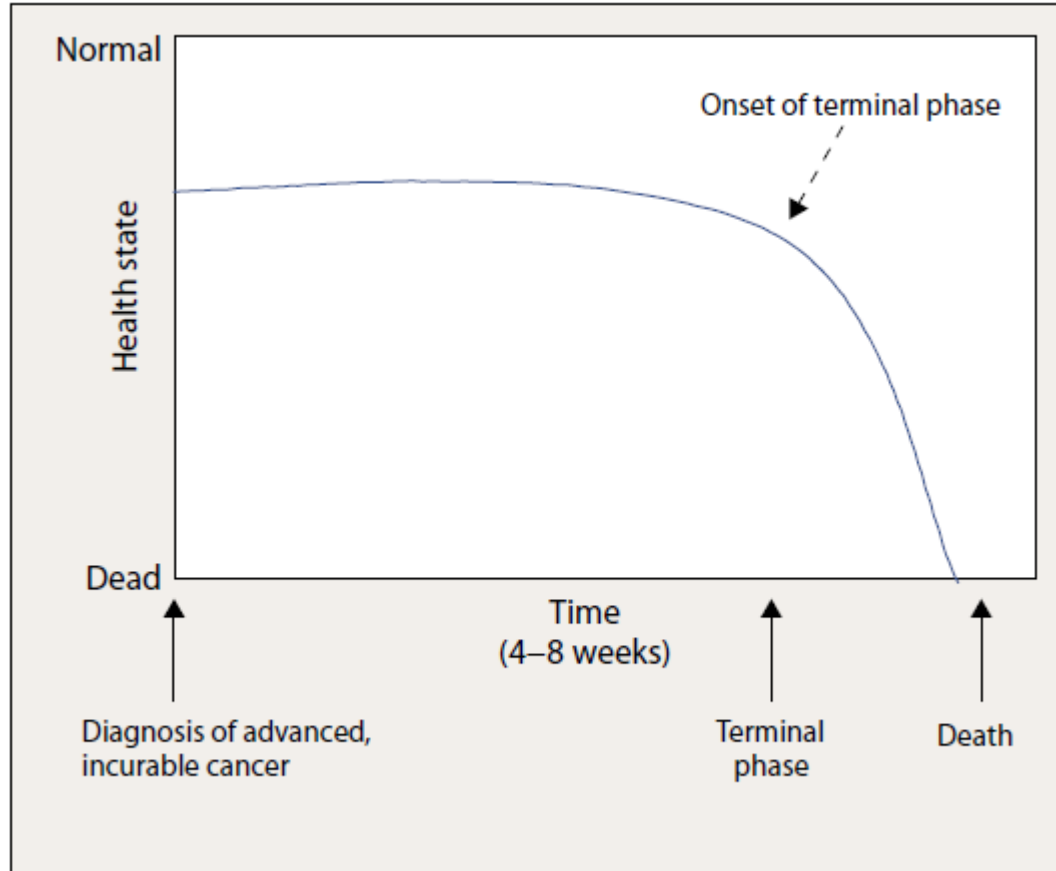
Cancer Death Rates

Change in US Death Rates* by cause, 1950 and 2014



*Age-adjusted to 2012 US standard population. Sources: 1950 Mortality Data – CDC/ NCHS, NVSS, Mortality Revised. 2012 Mortality Data – National Center for Health Statistics, Centers for Disease Control and Prevention, 2012. <http://www.cdc.gov>

Typical 'death trajectory' of patients with advanced cancer



“Terminal phase” is not the same as “Terminal illness” and insurers definition of “terminal illness” is different again (< 12 months life expectancy for insurer definition).

Growing availability of new oncological treatments will transform the trajectory of the advanced cancer into a chronic condition

Survival Estimates

- Survival estimates are critical factors in clinician and patient decision making in all phases of a serious and/or life-threatening illness
- Prognostic estimates have increased importance as they approach the end of life
- In practice, the most frequently employed method of prognosticating is the clinician prediction of survival (CPS)
- This relies on clinicians' experience and is subject to cognitive bias (often subjective systematic pattern of deviation from rationality in judgement)

How do we assess survival in clinic?

- Physicians are able to determine the general distribution of patient survivals
- The actual times predicted tend to be inaccurate, by an average of 2–3 months

How Physicians Untrained in Prognostication Might Formulate a Prognosis

Guess or use “intuition”
Ask an “expert” (oncologist, palliative care specialist)
Consult a textbook
Search the electronic literature for prognostic studies
Rely on own judgment
Use a prognostic index

Median survival can be translated into best, worst, or most likely outcomes -

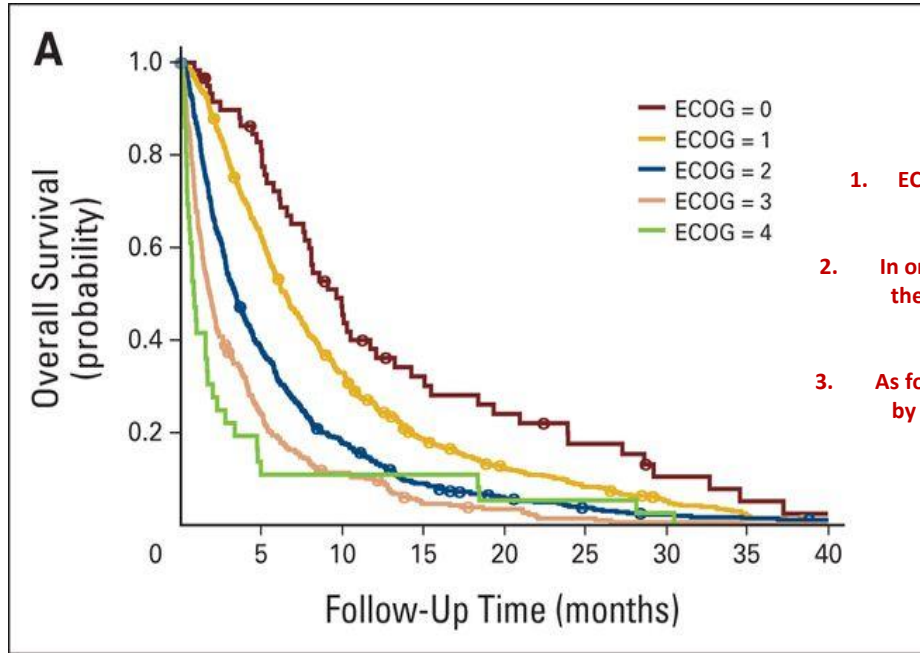
Table 2 Median Survival in Terms of the Most Likely, Best, and Worst Outcomes

Median Survival	Most Likely (1/2–2 × Median)	Best (3–4 × Median)	Worst (1/6th Median)
6 mo	3–12 mo	18–24 mo	1 mo
1 yr	6 mo–2 yr	3–4 yr	2 mo
2 yr	1–4 yr	6–8 yr	4 mo
3 yr	1.5–6 yr	9–12 yr	6 mo
4 yr	2–8 yr	12–16 yr	8 mo
5 yr	2.5–10 yr	15–20 yr	10 mo
6 yr	3–12 yr	18–24 yr	12 mo
7 yr	3.5–14 yr	21–28 yr	14 mo
8 yr	4–16 yr	24–32 yr	16 mo
9 yr	4.5–18 yr	27–36 yr	18 mo
10 yr	5–20 yr	30–40 yr	20 mo

Possible Prognostic Factors in Patients With Advanced Cancer

TUMOR-RELATED	PATIENT-RELATED	ENVIRONMENTAL
Anatomic extent	Age, gender, race	Marital status, social support
Histologic grade and other features	Genetic make up	Geographic location
Tumor bulk	Performance status	Access to treatment
Tumor marker level	Symptoms and signs	Expertise of oncologist
Tumor molecular biology	Psychologic status	Response to treatment
Disease-free interval	Function of vital organs	Oncologist's prediction of survival
	Comorbidities	

Survival curves for all patients by (A) Eastern Cooperative Oncology Group performance status (ECOG) and (B) Palliative Performance Scale score (PPS).



1. ECOG of 4 corresponds to a median survival of approximately 25 days
2. In order to easily calculate the median survivals for the other ECOG levels, which increase by 50% for each level.
3. As for any model, this estimate requires adjustment by the clinician, based on clinical impression and experience



Anxiety
Depression
Fatigue
Dyspnea
Pain
Infection
..... Many more

ASK FOR PERFORMANCE STATUS

TABLE Models for predicting prognosis in patients with advanced cancer

Study/model name	Patient population (N)	Factors in model (estimated life expectancy)	Strengths/weaknesses
Pirvano, 1999/Palliative Prognostic Score (PaP) ¹⁹	Advanced solid tumors no longer considered suitable for anti-cancer therapies and determined unlikely to live >6 months (N = 519)	Anorexia, dyspnea, KPS, total WBC, lymphopenia, physician's survival prediction in weeks (30 d)	External validation ⁴⁸ /Requires physician prediction of survival, limited applicability for longer survivals (>30 days), study population limited to advanced cancer patients no longer receiving anti-cancer therapies
Morita, 1999/Palliative Prognostic Index (PPI) ¹⁸	Terminally-ill cancer pts admitted to a palliative care unit and determined unlikely to live >6 months (N = 150)	Palliative performance score, oral intake, edema, dyspnea at rest, delirium (3 wk, 6 wk)	External validation ⁴⁹ /Limited applicability for longer term survival (>6 weeks), study population limited to advanced cancer patients admitted to a single palliative care unit
Chow, 2008/Number of Risk Factors (NRF) ¹⁵	Advanced cancer patients referred for palliative radiation therapy (N = 395)	Primary tumor, site of metastasis, KPS (9 wk, 26 wk, 60 wk)	External validation, ease of use (3 clinical factors)/Study population limited to those receiving palliative radiation therapy, limited applicability to patients with shorter survivals (<9 weeks)
Zhou, 2009/Chinese Prognostic Scale (ChPS) ¹⁶	Advanced cancer patients referred to a palliative home care service (N = 1019)	Weight loss, nausea, dysphagia, dyspnea, edema, cachexia, dehydration, gender, KPS, QOL (3 mo)	Large patient population/Not externally validated, study population limited to cancer patients in a palliative care home setting
Chiang, 2010 ⁵⁰	Terminal cancer patients in an inpatient hospice setting (N = 727)	Gender, intervention tubes, grade 3 edema, ECOG PS, mean muscle power, hemoglobin, BUN, SGOt, respiratory rate, heart rate (7 d)	Large patient population/No external validation, study population limited to cancer patients in a hospice setting, limited utility among patients with longer survivals (>7 days)
Hyodo, 2010 ⁵¹	Inpatient advanced cancer patients in cancer centers or hospice (N = 201)	Physicians' survival prediction (in weeks), consciousness, pleural effusion, WBC, lymphocyte % (2 wk, 5 wk, 7 wk)	Internal validation/No external validation, requires physician prediction of survival, limited utility among patients with longer survivals (>30 days), study population limited to inpatient advanced cancer patients
Martin, 2010 ⁵²	Patients with metastatic cancer referred to a palliative care center (N = 1164)	Primary tumor, PS, short-term weight change, dietary intake, dysphagia (1 mo, 2 mo, 4 mo)	Large patient cohort, internal validation/Not externally validated, study population limited to cancer patients in a palliative care center
Ohde, 2011 ⁵³	Terminally-ill cancer pts in a palliative care unit (N = 158)	Anorexia, dyspnea, edema, BUN, platelet count (2 wk)	Ease of use/Not externally validated, small study population from single palliative care site, limited utility among patients with longer survivals (>2 weeks)
Suh, 2010/Objective prognostic score (OPS) ²⁰	Terminally-ill cancer patients (N = 209)	Reduced oral intake, dyspnea at rest, ECOG performance status, leukocytosis, elevated bilirubin, elevated creatinine, elevated LDH (3 wk)	Multicenter study/Not externally validated, limited utility among patients with longer survivals (>3 weeks)
Feliu, 2011 ⁵⁴	Terminally-ill cancer pts in oncology and palliative care units no longer receiving anti-cancer therapies (N = 406)	ECOG PS, LDH, albumin, lymphocyte count, time from initial diagnosis to terminal cancer diagnosis (15 d, 3 d, 60 d)	External validation, ⁵⁴ ease of use/Limited utility among patients with longer survivals (>60 days), study population limited to patients no longer receiving anti-cancer therapies
Tredan, 2011 ²¹	Patients with locally advanced or metastatic cancer who had received at least one line of systemic therapy for metastatic disease (N = 299)	KPS, LDH, lymphocyte count, IL-6, albumin, platelets (4 mo, 5 mo, 18 mo)	Dichotomous variables used to make clinical interpretation easier and less subjective/Not externally validated, limited utility among patients with shorter survivals (<4 months)
Gwilliam, 2011/Prognosis in palliative care study (PiPSA) ⁵⁵	Patients with advanced, incurable cancers from palliative care centers (N = 1018)	Mental test score, pulse rate, presence of metastases, site of metastases, ECOG performance score, global health score, loss of appetite, dyspnea, dysphagia, primary cancer, weight loss (2-8 wk)	Large population from multiple centers, compared to accuracy of physician estimates/Not externally validated, study population limited to those seen at palliative care centers, limited utility among patients with longer survivals (>2 months)

Prognostic Factors by Percent of Models

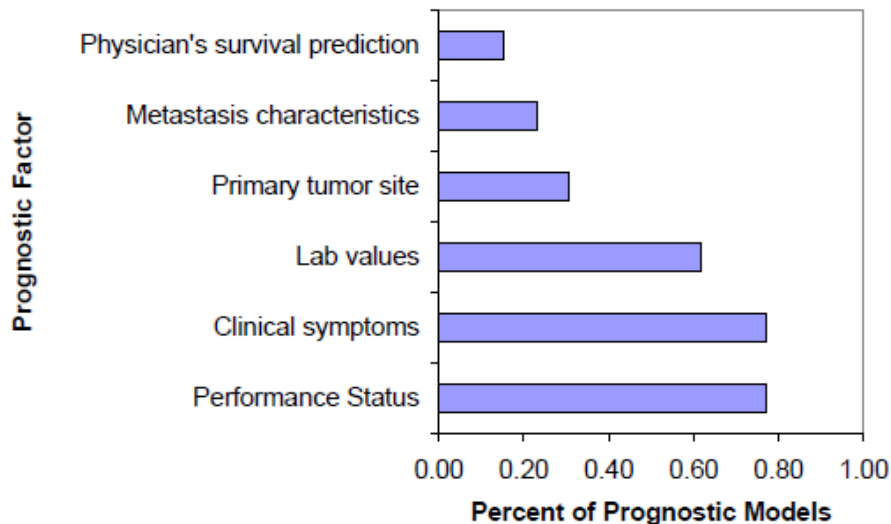


FIGURE Prognostic factors by proportion included in prognostic models.*

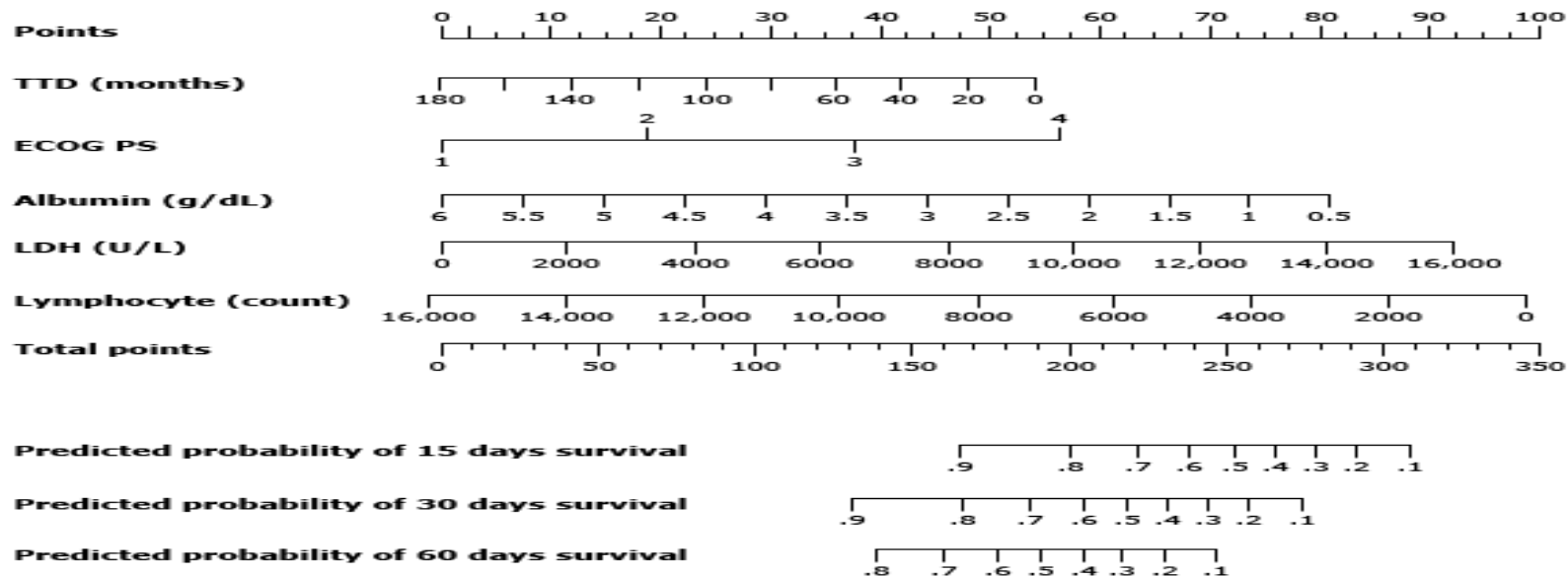
Actual survival Count of patients (% of total)					Absolute agreement between prediction and survival
	Days	Weeks	Months+	Total	
Doctors' prediction					
Days	137 (16.5)	16 (1.9)	3 (0.4)	156 (18.8)	56.3%
Weeks	82 (9.9)	126 (15.2)	86 (10.4)	294 (35.5)	
Months+	36 (4.3)	139 (16.8)	204 (24.6)	379 (45.7)	
Total	255 (30.8)	281 (33.9)	293 (35.3)	829 (100.0)	
Nurses' prediction					
Days	135 (14.2)	19 (2.0)	7 (0.7)	161 (16.9)	55.2%
Weeks	84 (8.8)	116 (12.2)	79 (8.3)	279 (29.2)	
Months+	52 (5.5)	186 (19.5)	276 (28.9)	514 (53.9)	
Total	271 (28.4)	321 (33.6)	362 (37.9)	954 (100.0)	
Combined MDT prediction					
Days	147 (14.9)	17 (1.7)	4 (0.4)	168 (17.0)	57.5%
Weeks	96 (9.7)	136 (13.8)	92 (9.3)	324 (32.8)	
Months+	35 (3.5)	176 (17.8)	284 (28.8)	495 (50.2)	
Total	278 (28.2)	329 (33.3)	380 (38.5)	987 (100.0)	
Patients' prediction					
Days	2 (0.7)	1 (0.3)	2 (0.7)	5 (1.7)	52.1%
Weeks	15 (5.2)	17 (5.9)	5 (1.7)	37 (12.8)	
Months+	30 (10.3)	86 (29.7)	132 (45.5)	248 (85.5)	

'Horizon effect' describes the phenomenon whereby clinicians appear to be more accurate at predicting survival when patients have a shorter time to live

- Survival estimates of clinicians:
significantly ($P < 0.001$) more likely to be optimistic (doctors: 31%) than pessimistic (12.7%).
- MDT prediction was correct
 - 87.5% of the time when predicting a survival of 'days'
 - 42.0% of the time when predicting a survival of 'weeks' and
 - 57.5% of the time when predicting a survival of 'months'.

<http://annonc.oxfordjournals.org/content/24/2/482.full>

Nomogram to predict the probability of survival at 15, 30, and 60 days in terminally ill cancer patients



1. Using a ruler or straight edge, draw a line from each of the following up to "Points" at the top:

- Time from initial diagnosis to diagnosis of terminal disease (TTD)
- Eastern Cooperative Oncology Group Performance Status (ECOG PS)
- Serum albumin levels
- Serum lactate dehydrogenase levels (LDH)
- Lymphocyte count

2. Sum the points for these factors to obtain "Total points".

3. Draw a line from "Total points" straight down to obtain predictions of 15, 30, and 60 days survival.

Retrospective studies confirm... less value in > 6 month predictions

- Analysed 1563 clinician predictions, have indicated that CPS has better predictive value in the short term but is less accurate in the longer term (i.e. beyond 6 months)
- CPS is generally overoptimistic (median CPS 42 days, median AS 29 days)
- Correct to within one week in 25% of cases and overestimated survival by at least four weeks in 27%.
- **The longer the CPS the greater the variability in Actual Survival**
- Estimates of the duration of survival of incurable patients were well calibrated, but individual predictions were imprecise. Discrimination between patients who would survive for three months, and those who would not, was only fair ($A_{ROC} = 0.75 \pm 0.04$), and discrimination between patients who would survive for one year and those who would not was very poor ($A_{ROC} = 0.57 \pm 0.01$)

Why strive for accurate prediction

- Accurate predictions of survival in patients with advanced cancer are important for several reasons:
- A prognosis of less than 6 months is used as a criterion for hospice admission
- A prediction is necessary to respond to the awkward question “Doctor, how much time do I have?”
- Prognosis may be important for research design and analysis; and
- Perhaps most importantly, clinical decision-making in patients with advanced disease depends on accurate predictions.

Goal of improved prognostication

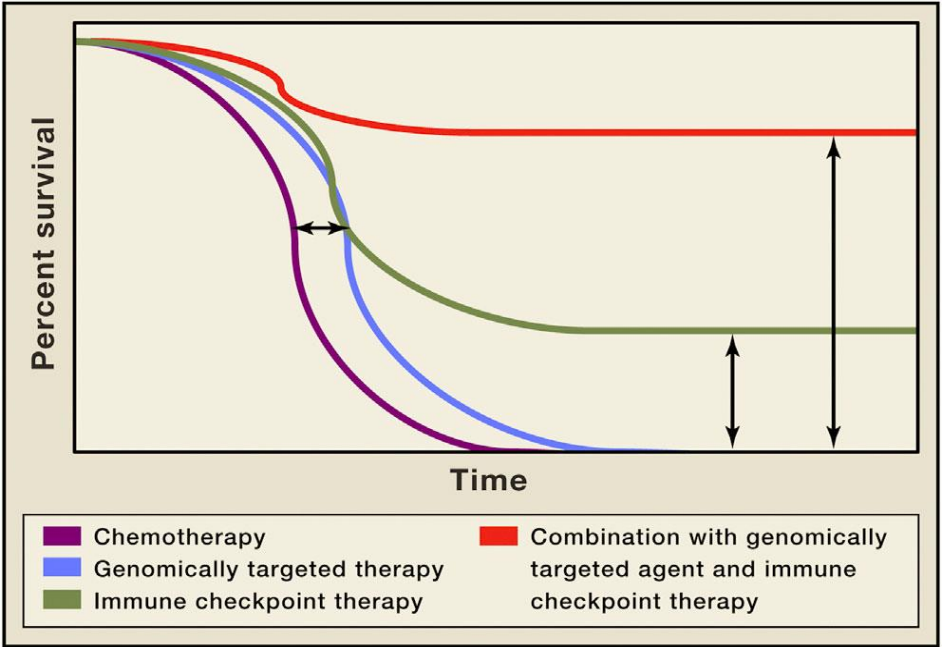
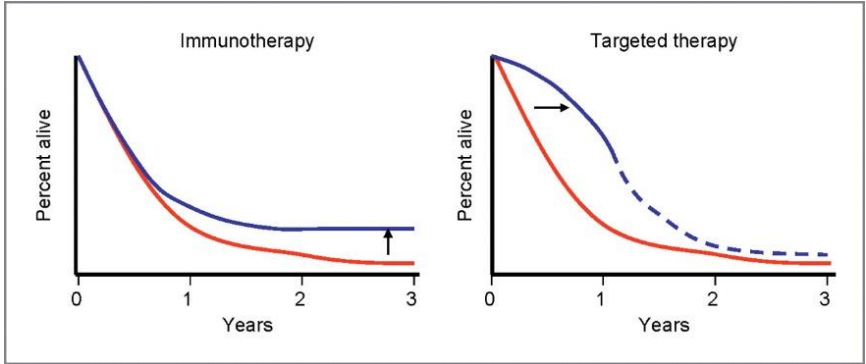
- The goal is to provide patients with a better understanding of their expected survival, thereby allowing them to make informed medical and social choices regarding their treatment path at the end of life, whether disease-directed, palliative, or a combination of both.

Agenda

- Challenges Oncologists face in accurately predicting survival
- **New Therapies**
 - Approved / non-approved
 - Personalised / targeted & Immunotherapy
- Cancers with greatest prognostic improvements
- Costs of clinical trials and gaining approval
 - Self funded
 - Postcode prescribing – is it a post code lottery?

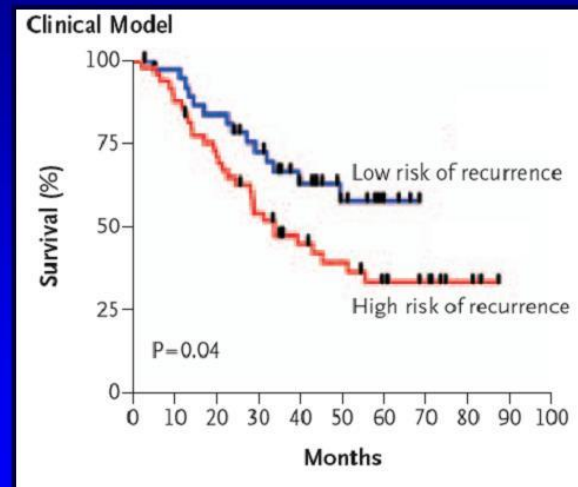
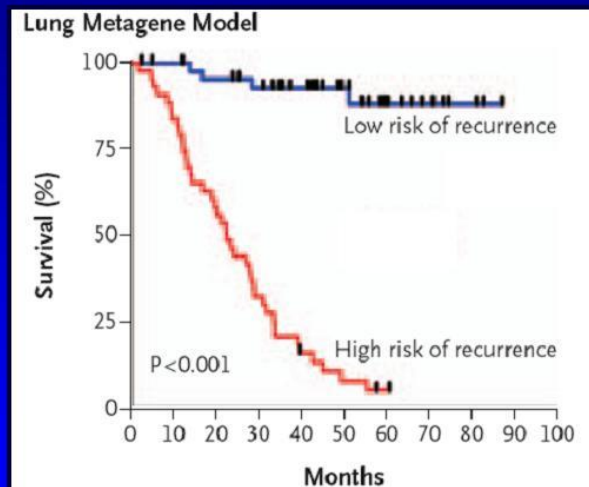
Two novel treatment modalities: TARGETED AND IMMUNOTHERAPY

Who brings what to the table?



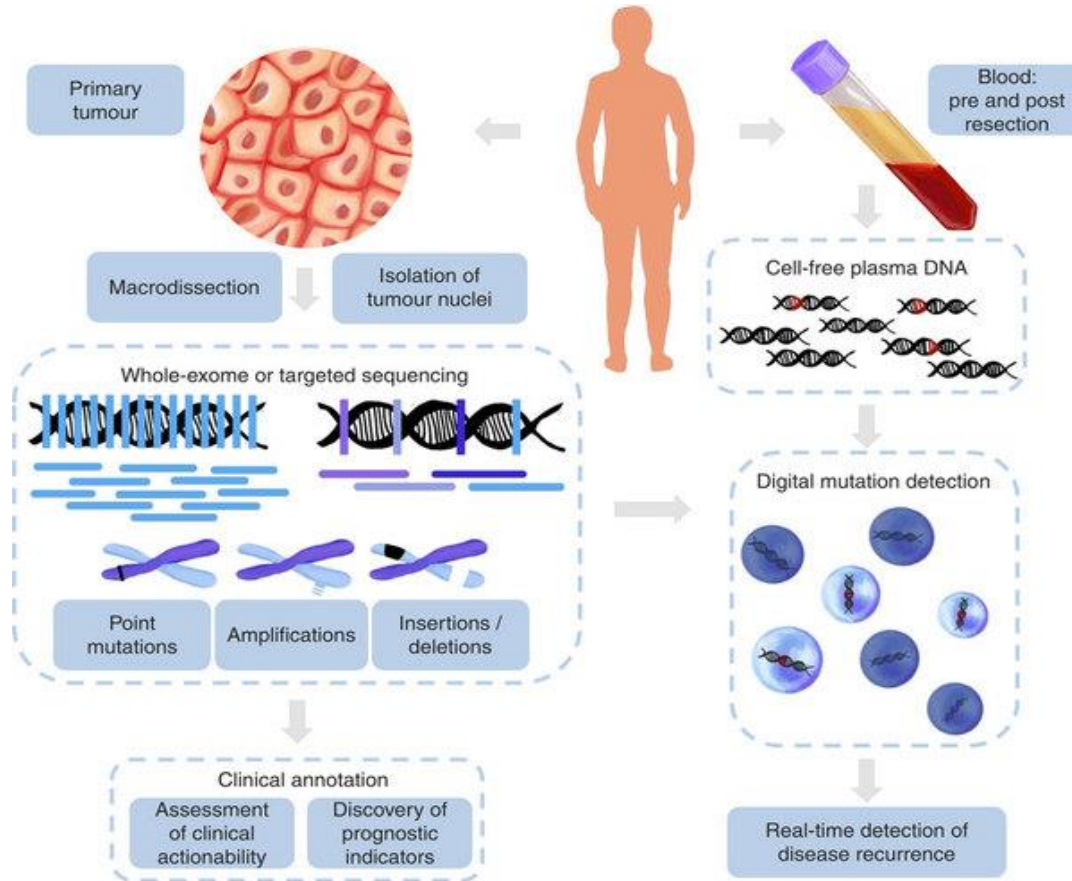
GENOMIC vs CLINICAL FACTORS: WHAT TIPS THE SCALE?

Survival Curves for Stage I Patient Population Based on Gene Expression vs. Clinical Factors

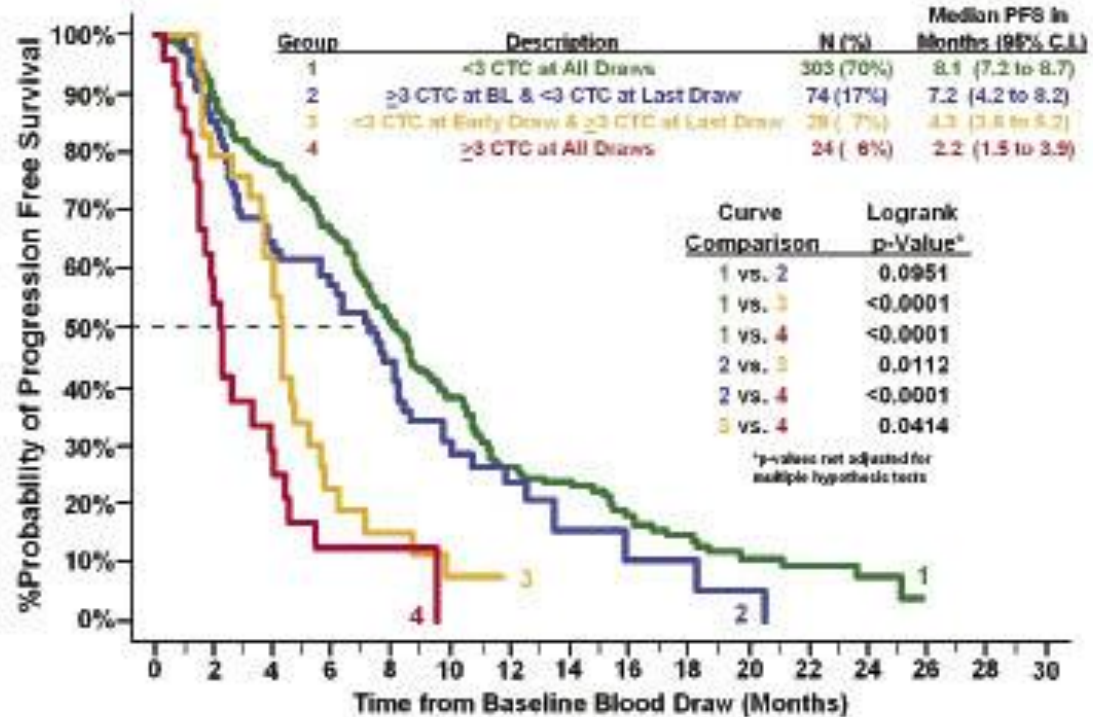
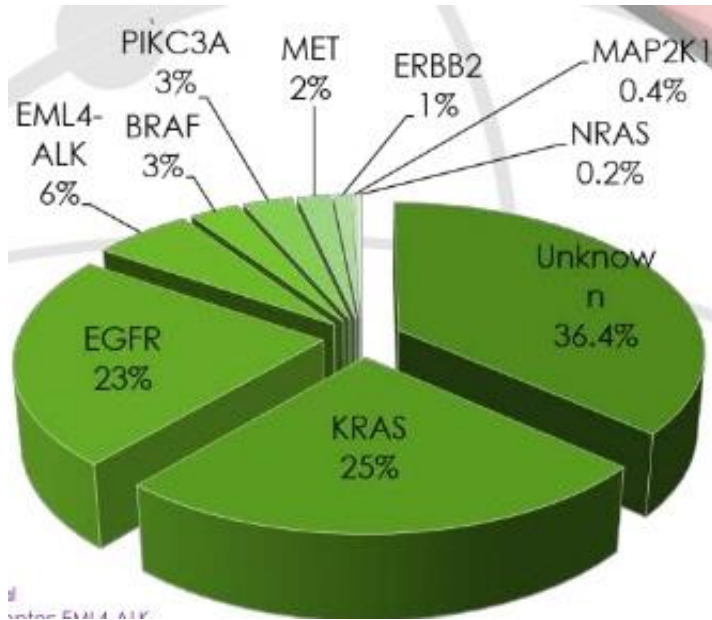


Potti et al., NEJM, 2006

What are we working towards?



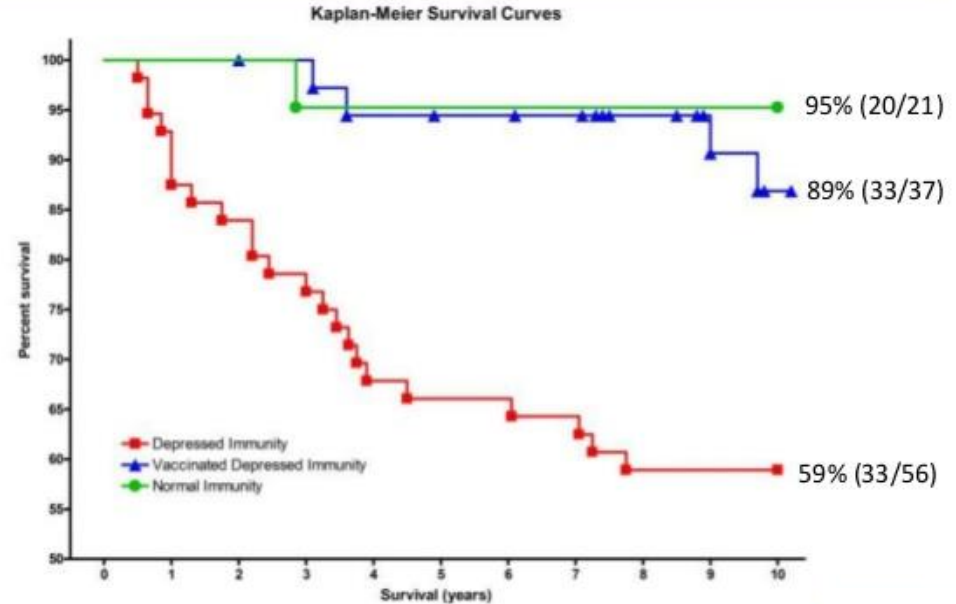
Tests we can do in clinic



CANCER AND IMMUNITY



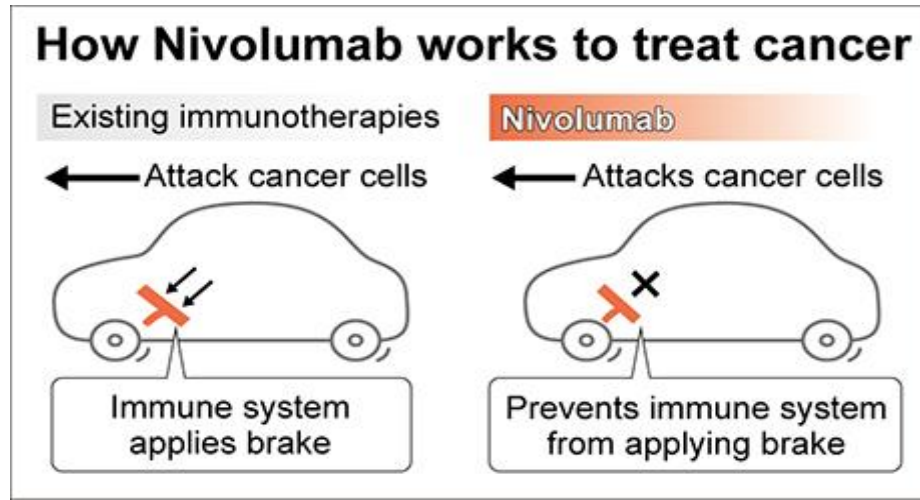
Kaplan-Meier Curve Shows Improved Survival of Breast Cancer patients with depressed immunity



Vaccination study using OBM vaccine technology

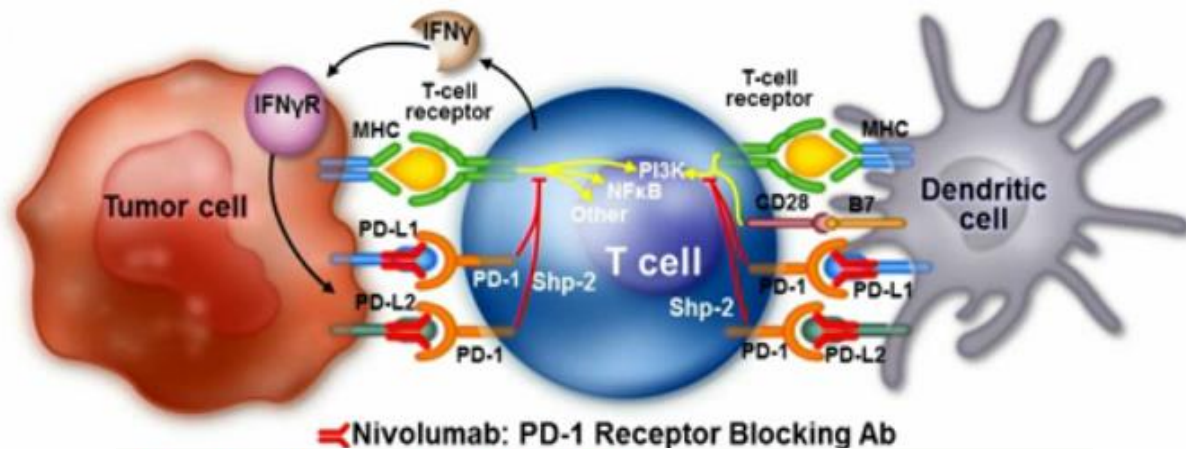
IMMUNOTHERAPY

- Immunotherapy teaches body to attack cancer cells and destroy tumours
- The treatment will replace chemotherapy within five years
- Tolerance of immunotherapy is far superior than that of chemotherapy
- Given in combination, IO does not potentiate side effects of other drugs
- It is particularly effective against skin and lung cancer, experts believe
- Patients expected to only survive months go on to live normal lives



Block PD1----T cell Activation

PD-1 blockade as a strategy for cancer immunotherapy^{2,4}



Pembrolizumab (MK-3475)

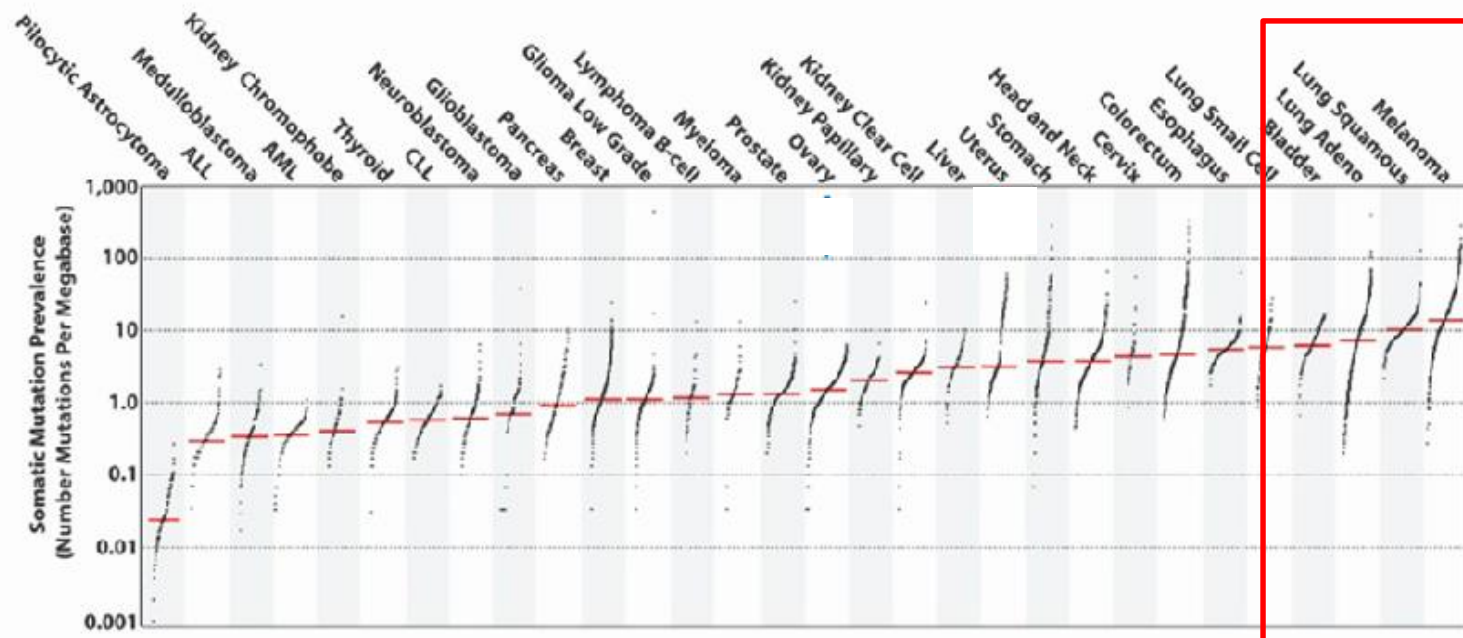
Nivolumab Mechanism of Action

- PD-1 expression on tumor-infiltrating lymphocytes is associated with decreased cytokine production and effector function¹¹
- Nivolumab binds PD-1 receptors on T cells and disrupts negative signaling triggered by PD-L1/PD-L2 to restore T-cell antitumor function¹²⁻¹⁴
 - Dual blockade of PD-L1 and PD-L2
 - No cytotoxic (ADCC/CDC) activity

Agenda

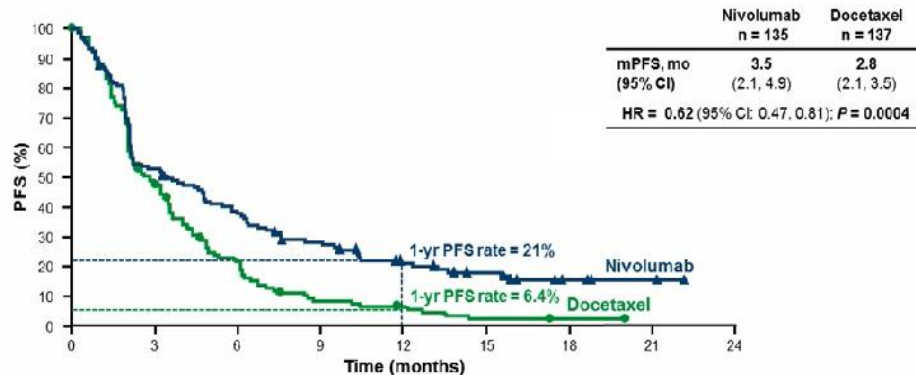
- Challenges Oncologists face in accurately predicting survival
- New Therapies
 - Approved / non-approved
 - Personalised / targeted & Immunotherapy
- **Cancers with greatest prognostic improvements**
- Costs of clinical trials and gaining approval
 - Self funded
 - Postcode prescribing – is it a post code lottery?

Tumour mutational load and suitability for immunotherapy



Alexandrov et al, Nature, 2103, 500: 415

Progression-Free Survival

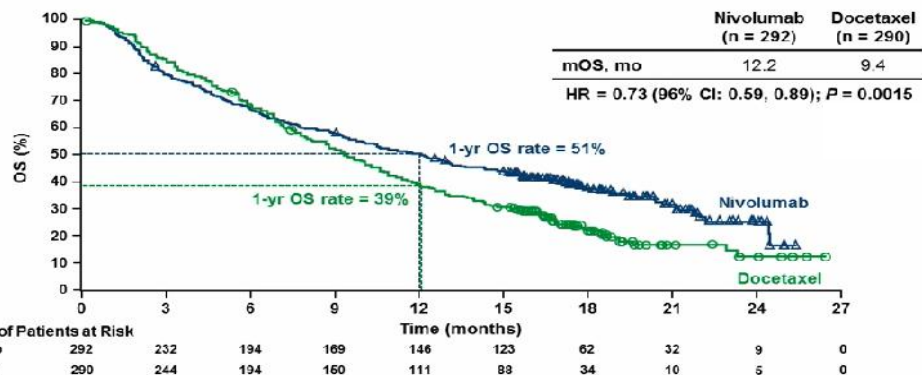


CHECKMATE NON SMALL CELL LUNG CANCER

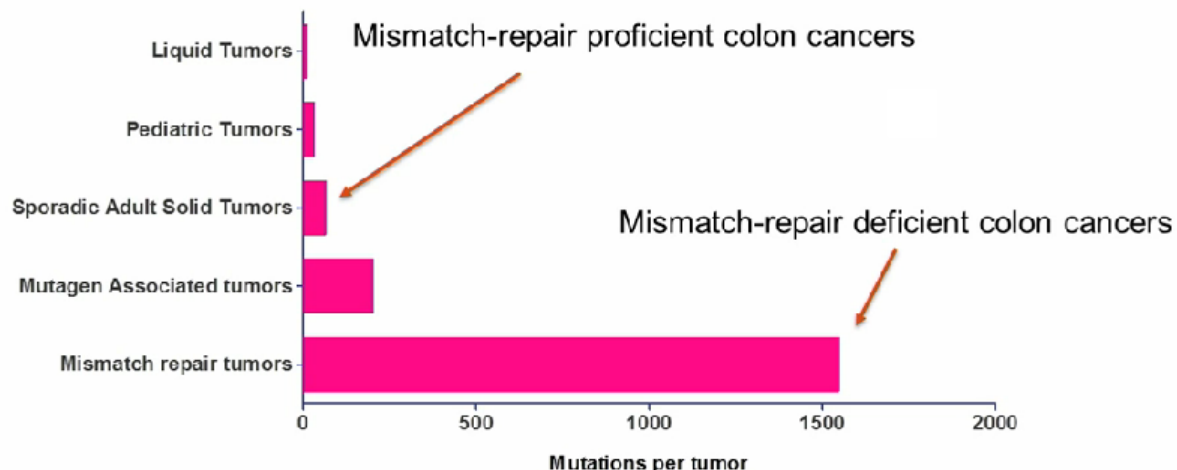
Number of Patients at Risk

Nivolumab	135	68	48	33	21	15	6	2	0
Docetaxel	137	62	26	9	6	2	1	0	0

Overall Survival



Mutations per tumor

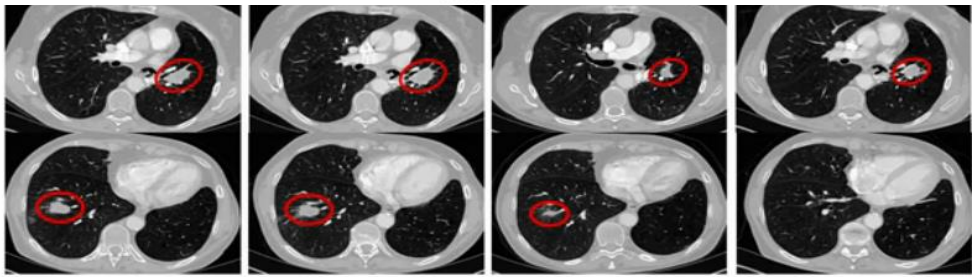
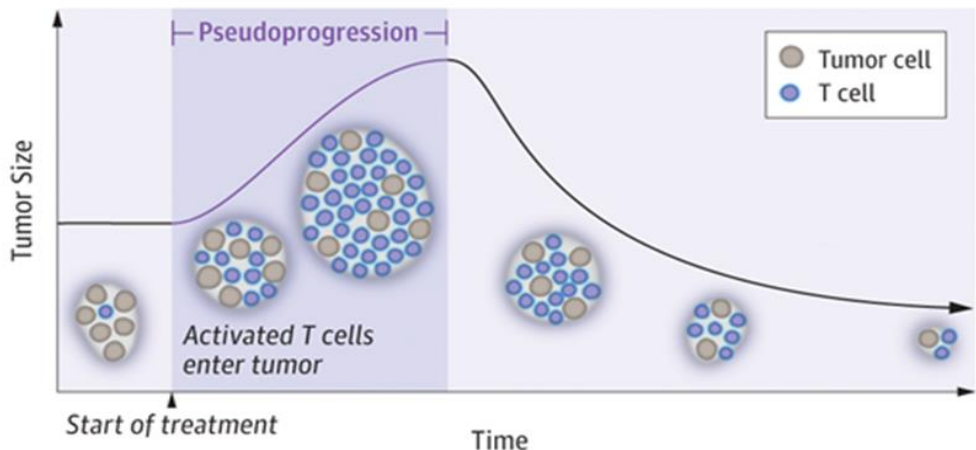


Objective Responses

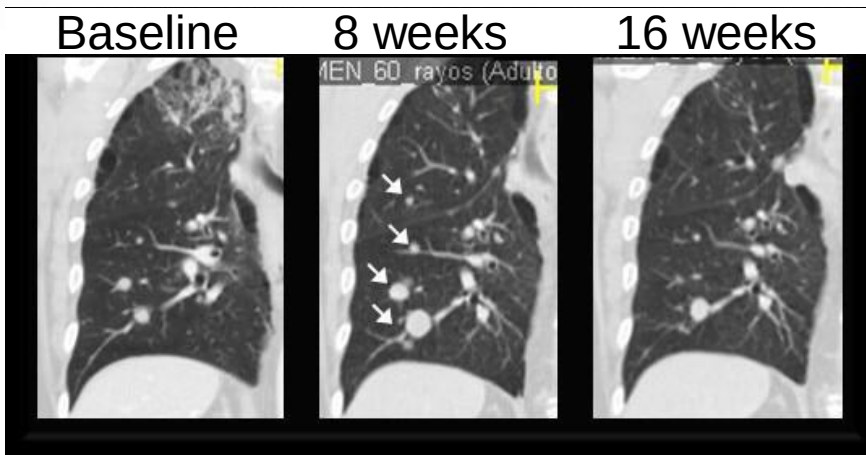
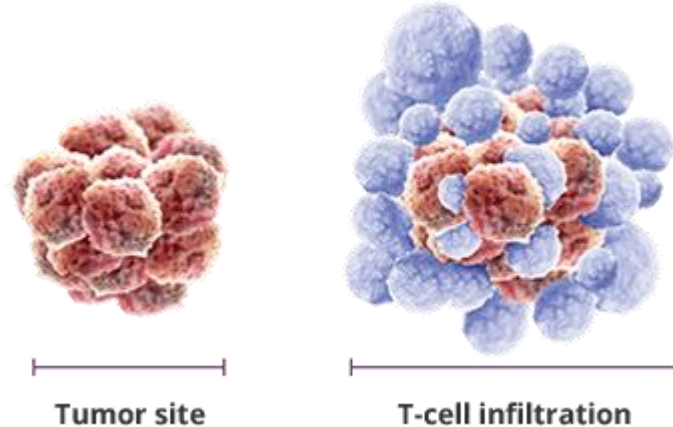
	MMR-deficient CRC	MMR-proficient CRC	MMR-deficient non-CRC
<i>N</i>	13	25	10
Objective Response Rate	62%	0%	60%
Disease Control Rate	92%	16%	70%

Atypical Patterns of Response in Patients

Response to immune checkpoint inhibitor treatment with brief increase in tumor size (pseudoprogression)



Baseline 9 wks 18 wks 27 wks



ASK FOR irRECIST review



Role of Nivolumab in Hodgkin Lymphoma

- Nivolumab produced a response in 87% of patients.
- Responses were observed in patients with failure of both stem cell transplantation and brentuximab vedotin, those with failure of brentuximab and no prior stem cell transplantation, and those who received no prior brentuximab.

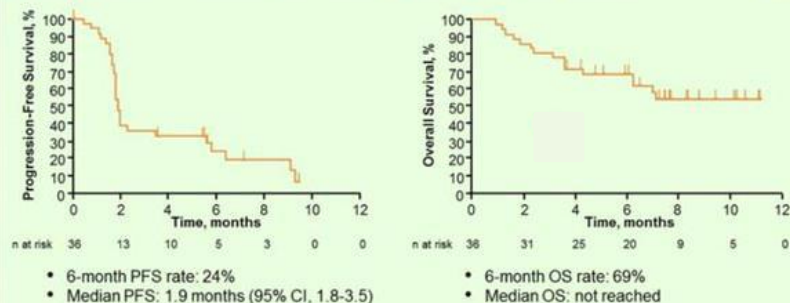
Preliminary Anti-Tumor Response and Overall Survival With Nivolumab

	Uninfected (n=21)	HCV (n=11)	HBV (n=10)	Total Patients Evaluable (n=42)
Objective response rate, n (%)	3 (14)	4 (36)	1 (10)	8 (19)
Complete response	2 (10)	0	0	2 (5)
Partial response	1 (5)	4 (36)	1 (10)	6 (14)
Ongoing response, n (%)	3/3 (100)	3/4 (75)	0	6/8 (75)

Pembrolizumab Findings Spark Interest

- The anti-PD-1 agent pembrolizumab produced an 18.5% response rate in patients with metastatic triple-negative breast cancer.
- For responders, median duration of response has not been reached; 6-month progression-free survival was 23.3%.

Kaplan-Meier Estimates of Survival



Analysis cut-off date: November 10, 2014.

Presented at the Gastrointestinal Cancers Symposium

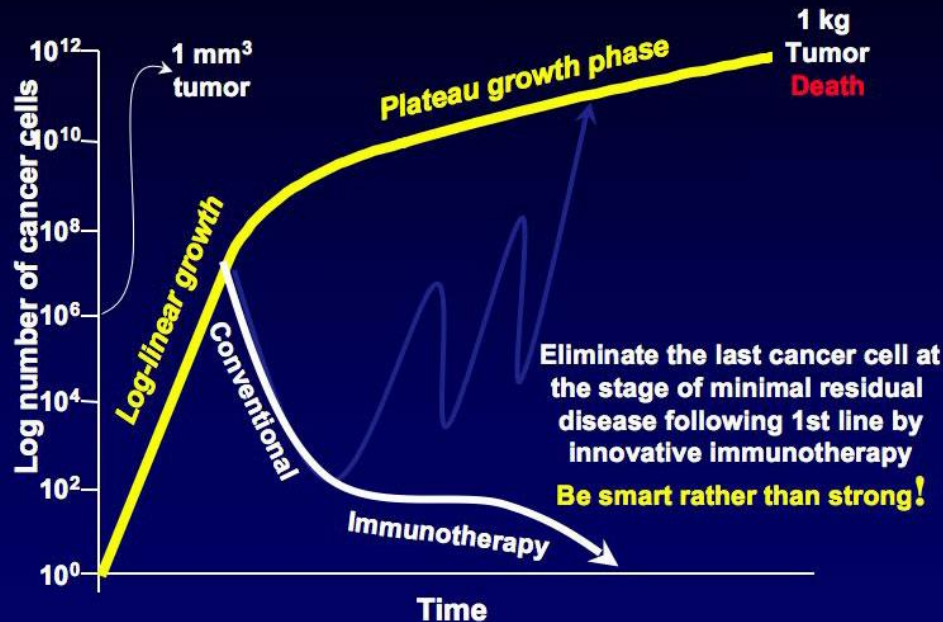
Presented by: Kei Muro

Early Findings with Pembrolizumab, Merck's Investigational Anti-PD-1 Therapy, in Patients with Advanced Pleural Mesothelioma Presented at AACR Annual Meeting

Pembrolizumab Demonstrated 28 Percent Overall Response Rate and 76 Percent Disease Control Rate in Difficult-to-Treat Cancer

Make the newest, game-changing drugs available EARLY

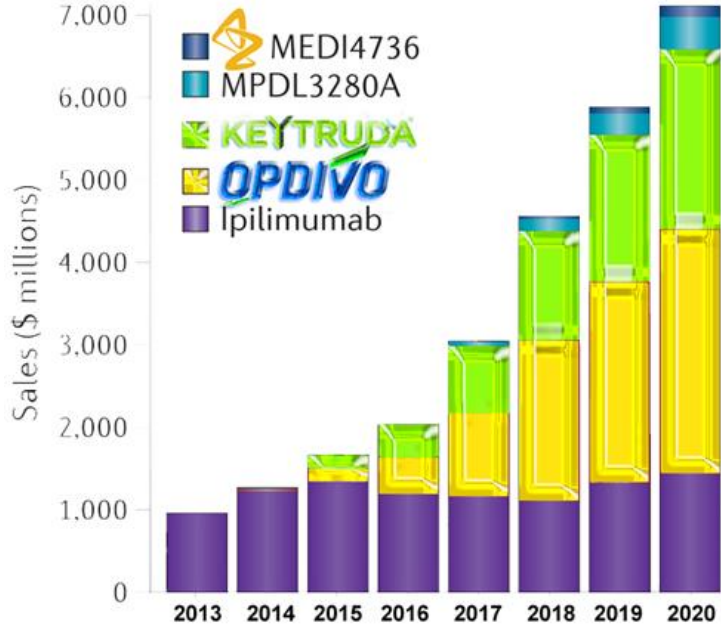
Cancer immunotherapy at the stage of minimal residual disease induced by conventional treatment at ICTC



1. We are learning who benefits most from IO – biomarkers ?
2. When best to start ?
3. Optimal Rx duration ?
4. Best combinations ?
5. Survival benefit persists after cessation of Rx !

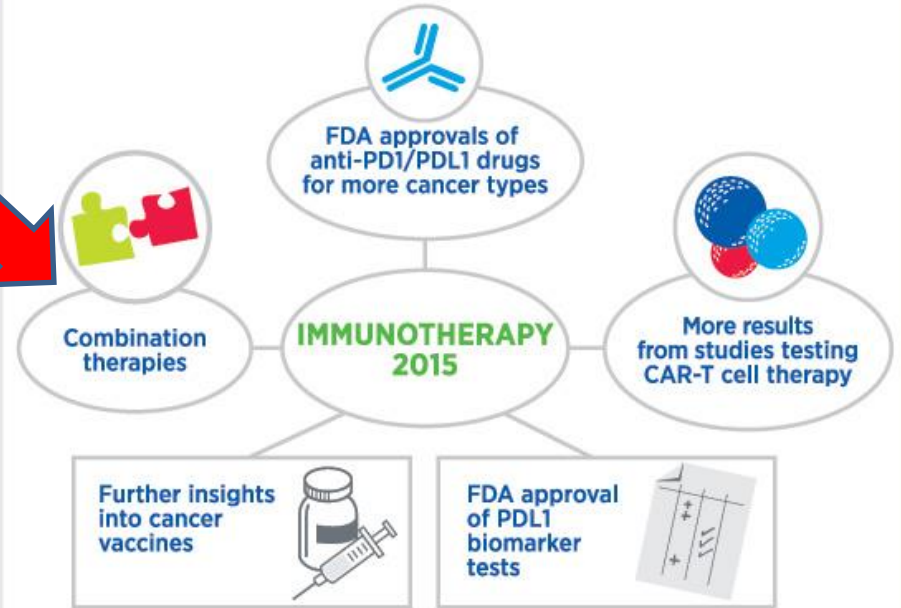
Agenda

- Challenges Oncologists face in accurately predicting survival
- New Therapies
 - Approved / non-approved
 - Personalised / targeted & Immunotherapy
- Cancers with greatest prognostic improvements
- **Costs of clinical trials and gaining approval**
 - Self funded
 - Postcode prescribing – is it a post code lottery?



Nature Reviews | Drug Discovery

WHAT'S DOWN THE ROAD FOR IMMUNOTHERAPY?



Graphic developed based on conversations with Pardoll

AACR American Association for Cancer Research

LUCID

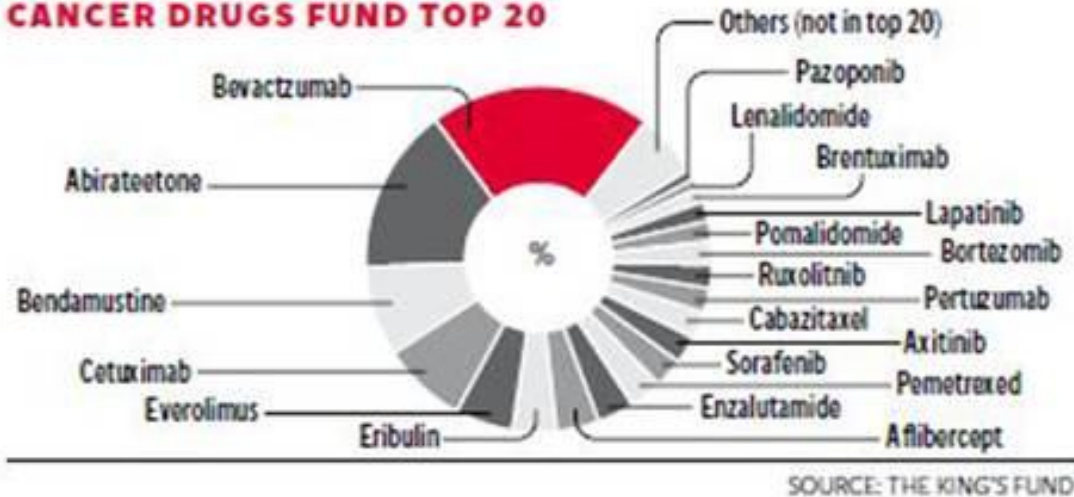
Immunotherapy

Cost and Access

- Yearly cost of immunotherapy is > 50,000 GBP
- Anti-CTLA4 and anti-PD1 are licenced for limited indications (melanoma, NCSLC)... awaiting for anti-PDL1 shortly
- Private health insurances slow to cover cost
- Some patients self-pay (linked with issues of prescribing, monitoring and follow up)
- Prices will reduce as these therapies gain acceptance and regulatory approval
- Access is often limited hence survival benefit is not delivered equally to all patients



CANCER DRUGS FUND TOP 20



At the beginning of 2015, there were 84 funded therapies, but after a series of culls there are now just 41. In the latest reduction, 23 courses of treatment have been removed. ‘

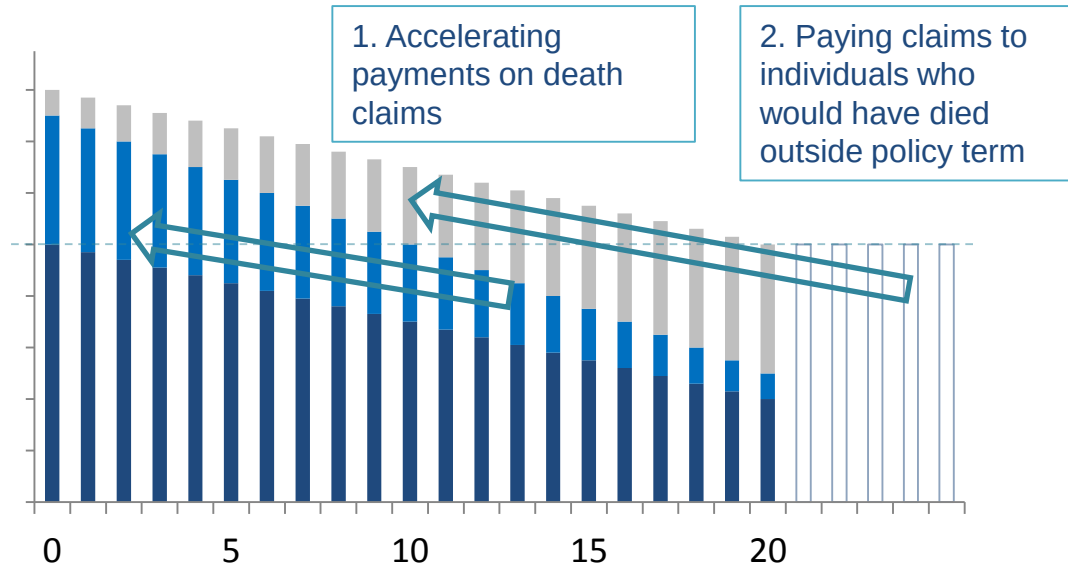
Conclusions

- Drawing upon clinical experience and comprehensive knowledge of the patient—is frequently utilized to assess prognosis
- Largely unreliable - with accuracy ranging from 20%-60%
- We need to develop more precise algorithms incorporating biological variables etc.
- Prediction beyond 6 months are largely inaccurate
- Treatment landscape of cancer is evolving at an unprecedented rate
- Patients that were considered terminal are now living longer on immunotherapy agents
- Our future is immunotherapy as the backbone of cancer treatment with support of chemotherapy and other treatment modalities
- **REALITY IS THAT CANCER WILL BECOME FOR MANY PATIENTS THE DISEASE THEY LIVE WITH
AND NOT THE ONE THEY DIE FROM**

Questions ?

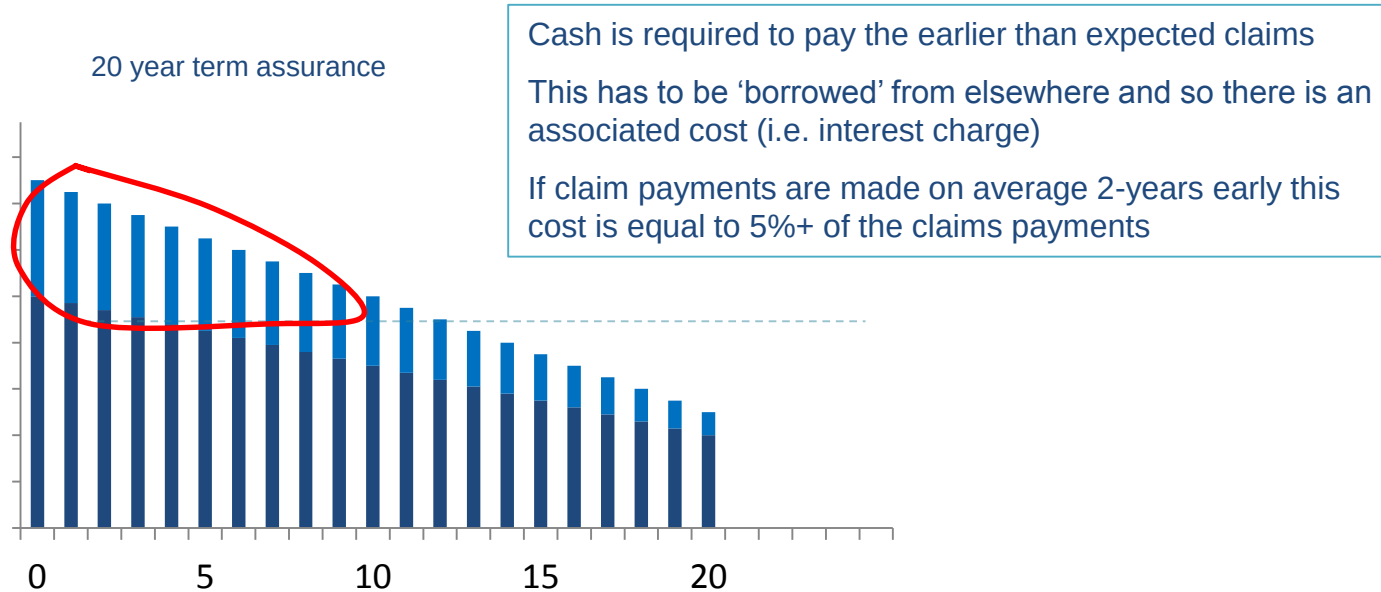
How does the presence of a terminal illness benefit affect the claims we pay?

20 year term assurance



Two components to the cost of offering a terminal illness benefit

Why does simply accelerating a claim payment cost anything at all?



Average acceleration of 1-2 years will materially affect profitability levels